CLOUD COMPUTING

UNIT-4

Cloud Computing

What is Cloud?

- The term **Cloud** refers to software and services that run on the internet, instead of locally on the user computer.
- Applications such as **e-mail, web conferencing, customer relationship management, DropBox** all run in cloud.

What is Cloud Computing?

- **Cloud Computing** refers to **manipulating, configuring,** and **accessing** the applications online. It offers online data storage, infrastructure and application.

WHY CLOUD COMPUTING

- Traditional business applications have always been very complicated and expensive. The amount and variety of hardware and software required to run them are daunting.
- Whole team of experts are needed to install, configure, test, run, secure, and update them.
- When you multiply this effort across dozens or hundreds of apps, it's easy to see why the biggest companies with the best IT departments aren't getting the apps they need.
- Small and mid-sized businesses don't stand a chance.

- **With Cloud:**
- With cloud computing, you eliminate those headaches because you're not managing hardware and software.
- The shared infrastructure means it works like a utility: You only pay for what you need, upgrades are automatic, and scaling up or down is easy.
- Cloud-based apps can be up and running in days or weeks, and they cost less. With a cloud app, you just open a browser, log in, customize the app, and start using it.

BENEFITS

- **COST**
- Eliminates the capital expence of buying h/w and s/w and installation expenses.
- SPEED
- Vast amounts of computing resources can be provisioned in minutes.
- GLOBAL SCALE
- Ability toscale elastically.(deleivering right amount of IT resources when it is needed and from right geographic locations)
- PRODUCTIVITY
- Eliminating h/w setup and s/w patching and other time consming IT management chores.So more time on achieving important bussiness goals.

BENEFITS

- **PERFORMANCE**
- Cloud computing services run on a worldwidenetwork of securedatacenters, which regularly upgraded to the latest generation of fast and efficient computing hardware.
- RELIABILITY
- Cloud computing makes data backup, disaster recovery and bussiness continuity easier and less expensive because data can be mirrored at multiple redudant sites on the cloud providers network.
- SECURITY
- Many cloud providers offer a broad set of policies,technology and controls that stengthen the security inturn helps in protecting the apps infrastructure and data from potensial threats.

HISTORY

- The concept of **Cloud Computing** came into existence in 1950 with implementation of mainframe computers, accessible via **thin/static clients**.
- Since then, cloud computing has been evolved from static clients to dynamic ones from software to s
- Before emerging the cloud computing, there was Client/Server computing which is basically a centralized storage in which all the software applications, all the data and all the controls are resided on the server side.

- If a single user wants to access specific data or run a program, he/she need to connect to the server and then gain appropriate access, and then he/she can do his/her business.
- Then after, distributed computing came into picture, where all the computers are networked together and share their resources when needed.
- On the basis of above computing, there was emerged cloud computing concepts.

- At around in 1961, John MacCharty suggested in a speech at MIT that computing can be sold like a utility, just like a water or electricity.
- But of course time has passed and the technology caught that idea and after few years we mentioned that:

- *In 1999,* **Salesforce.com** *started delivering of applications to users using a simple website*. The applications were delivered to enterprises over the Internet, and this way the dream of computing sold as utility were true.

- *In 2002,* **Amazon** *started Amazon Web Services*, providing services like storage, computation and even human intelligence. However, only starting with the launch of the Elastic Compute Cloud in 2006 a truly commercial service open to everybody existed.

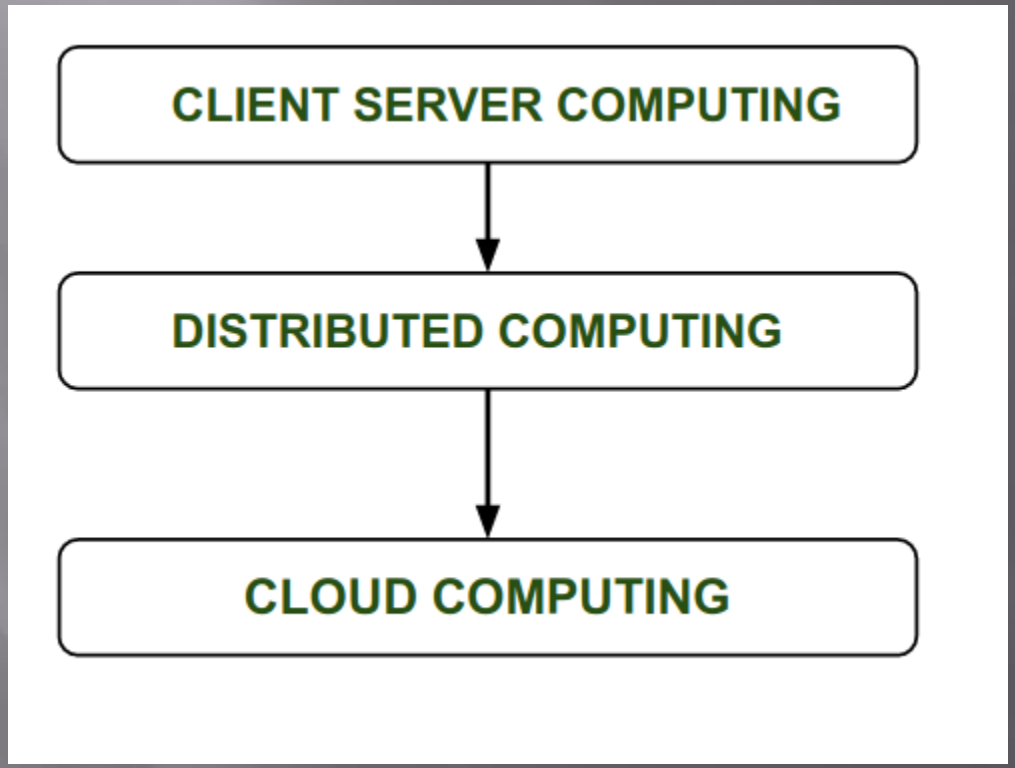
- *In 2009,* **Google Apps** *also started to provide cloud computing enterprise applications.*

- Of course, all the big players are present in the cloud computing evolution, some were earlier, some were later. *In 2009,* **Microsoft** *launched Windows Azure*, and companies like Oracle and HP have all joined the game.

S.No.	CLOUD COMPUTING	DISTRIBUTED COMPUTING
01.	Cloud computing refers to providing on demand IT resources/services like server, storage, database, networking, analytics, software etc. over internet.	Distributed computing refers to solve a problem over distributed autonomous computers and they communicate between them over a network.
02.	In simple cloud computing can be said as a computing technique that delivers hosted services over the internet to its users/customers.	In simple distributed computing can be said as a computing technique which allows to multiple computers to communicate and work to solve a single problem.

03.	**There are many benefits of cloud computing like cost effective, elasticity and reliable, economies of Scale, access to the global market etc.**	**There are many benefits of distributed computing like flexibility, reliability, improved performance etc.**
04.	Cloud computing provides services such as hardware, software, networking resources through internet.	Distributed computing helps to achieve computational tasks more faster than using a single computer as it takes a lot of time.
05.	The goal of cloud computing is to provide on demand computing services over internet on pay per use model.	The goal of distributed computing is to distribute a single task among multiple computers and to solve it quickly by maintaining coordination between them.

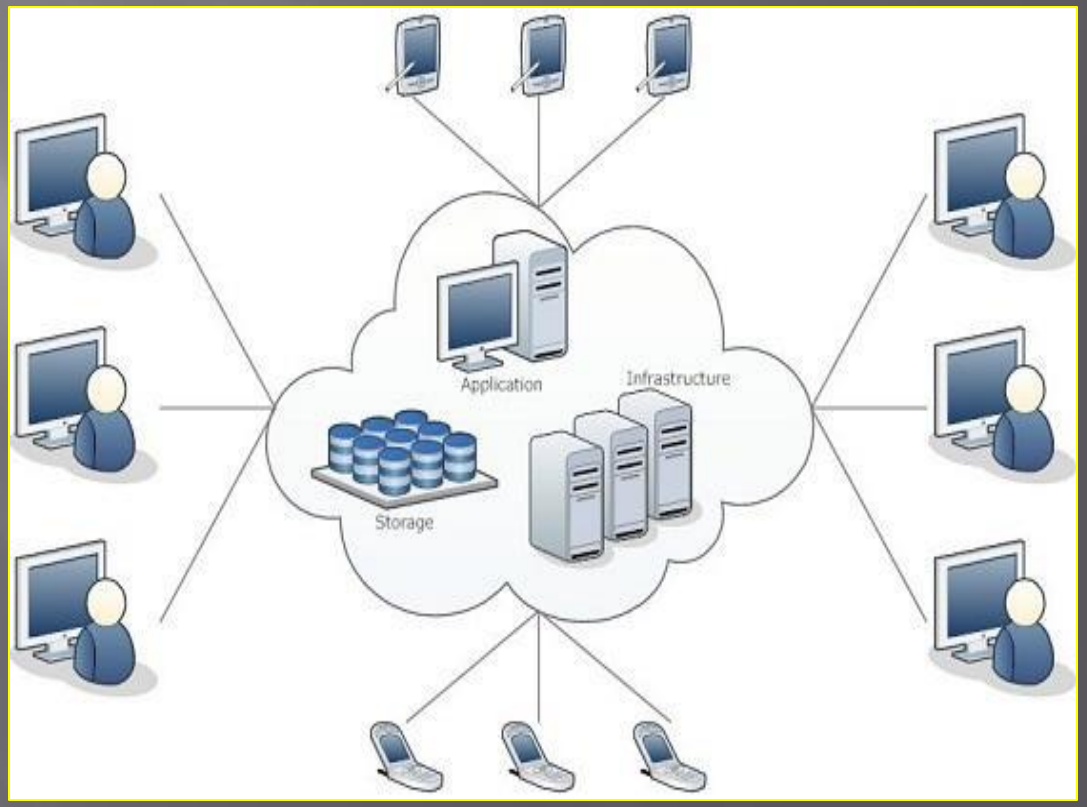
07.	Some disadvantage of cloud computing includes less control especially in the case of public clouds, restrictions on available services may be faced and cloud security.	Some disadvantage of cloud computing includes chances of failure of nodes, slow network may create problem in communication.


CLIENT SERVER COMPUTING
DISTRIBUTED COMPUTING
CLOUD COMPUTING

Basic Concepts

There are certain services and models working behind the scene making the cloud computing feasible and accessible to end users. Following are the working models for cloud computing:

- Deployment Models
- Service Models



DEPLOYMENT MODELS

- Deployment models define the type of access to the cloud, i.e., how the cloud is located? Cloud can have any of the four types of access: Public, Private, Hybrid and Community.

PUBLIC CLOUD

- The **Public Cloud** allows systems and services to be easily accessible to the general public. Public cloud may be less secure because of its openness, e.g., e-mail.

PRIVATE CLOUD

- The **Private Cloud** allows systems and services to be accessible within an organization. It offers increased security because of its private nature.

COMMUNITY CLOUD

- The **Community Cloud** allows systems and services to be accessible by group of organizations.

HYBRID CLOUD

- The **Hybrid Cloud** is mixture of public and private cloud. However, the critical activities are performed using private cloud while the non-critical activities are performed using public cloud.

SERVICE MODELS

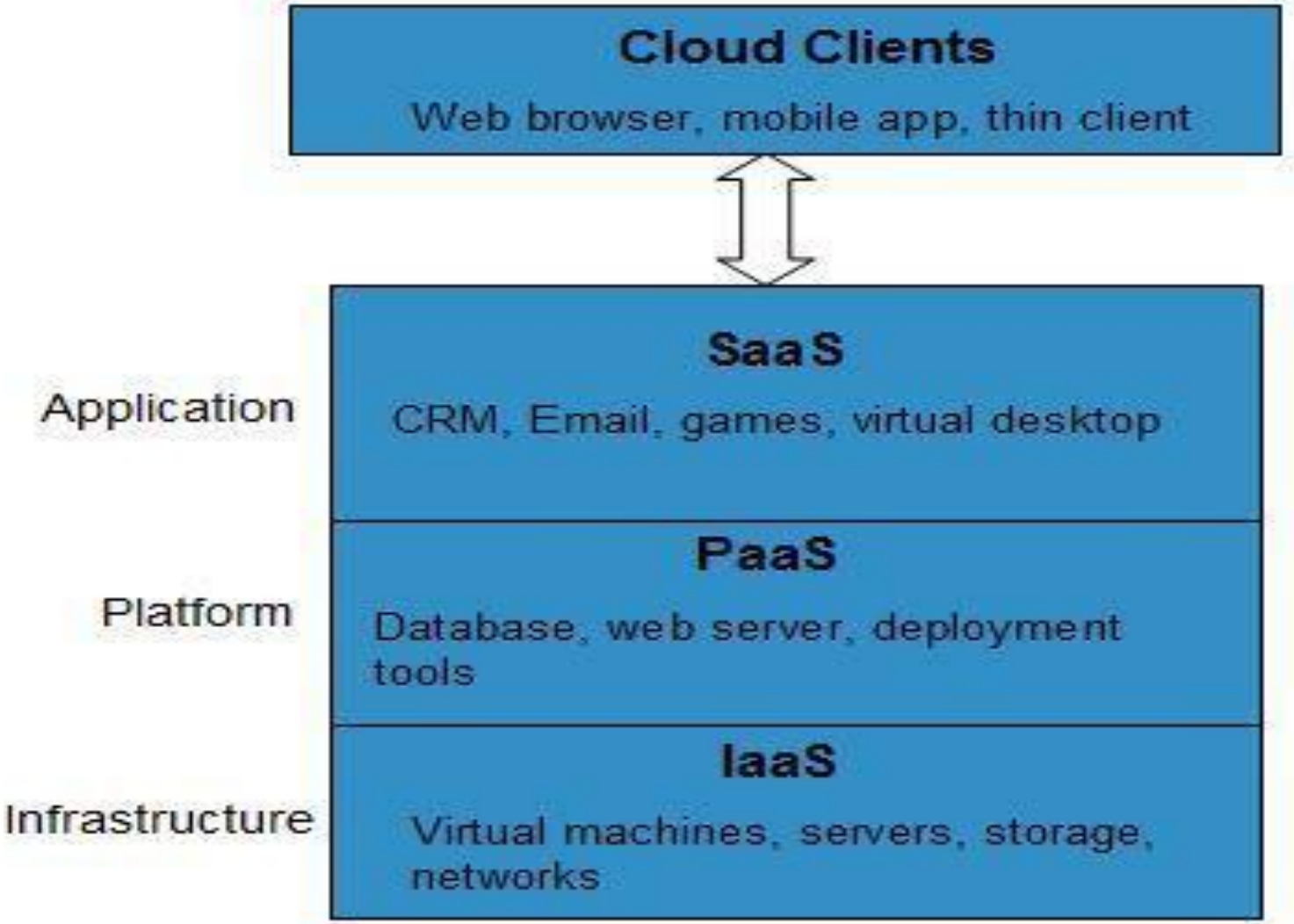
Service Models are the reference models on which the Cloud Computing is based. These can be categorized into three basic service models as listed below:

- 1. Infrastructure as a Service (IaaS)
- 2. Platform as a Service (PaaS)
- 3. Software as a Service (SaaS)

- There are many other service models all of which can take the form like **XaaS**, i.e., **Anything as a Service**. This can be **Network as a Service, Business as a Service, Identity as a Service, Database as a Service** or **Strategy as a Service.**

- The **Infrastructure as a Service (IaaS)** is the most basic level of service. Each of the service models make use of the underlying service model, i.e., each inherits the security and management mechanism from the underlying model, as shown in the following diagram:

SERVICE MODELS



SERVICE MODELS

INFRASTRUCTURE AS A SERVICE (IAAS)

- **IaaS** provides access to fundamental resources such as physical machines, virtual machines, virtual storage, etc.

PLATFORM AS A SERVICE (PAAS)

- **PaaS** provides the runtime environment for applications, development & deployment tools, etc.

SOFTWARE AS A SERVICE (SAAS)

- **SaaS** model allows to use software applications as a service to end users.

RISKS

Although Cloud Computing is a great innovation in the world of computing, there also exist downsides of cloud computing. Some of them are discussed below:

SECURITY & PRIVACY

- It is the biggest concern about cloud computing. Since data management and infrastructure management in cloud is provided by third-party, it is always a risk to handover the sensitive information to such providers. Although the cloud computing vendors ensure more secure password protected accounts, any sign of security breach would result in loss of clients and businesses.

LOCK-IN

- It is very difficult for the customers to switch from one **Cloud Service Provider (CSP)** to another. It results in dependency on a particular CSP for service.

RISKS

ISOLATION FAILURE

- This risk involves the failure of isolation mechanism that separates storage, memory, routing between the different tenants.

MANAGEMENT INTERFACE COMPROMISE

- In case of public cloud provider, the customer management interfaces are accessible through the Internet.

INSECURE OR INCOMPLETE DATA DELETION

- It is possible that the data requested for deletion may not get deleted. It happens either because extra copies of data are stored but are not available or disk destroyed also stores data from other tenants.

SERVICE MODELS

There are the following three types of cloud service models -

- Infrastructure as a Service (IaaS)
- Platform as a Service (PaaS)
- Software as a Service (SaaS)



INFRASTRUCTURE AS A SERVICE

IaaS is also known as **Hardware as a Service (HaaS)**. It is a computing infrastructure managed over the internet. The main advantage of using IaaS is that it helps users to avoid the cost and complexity of purchasing and managing the physical servers.

- **Characteristics of IaaS**

Key features:

- Instead of purchasing hardware outright, users pay for IaaS on demand.
- Infrastructure is scalable depending on processing and storage needs.
- Saves enterprises the costs of buying and maintaining their own hardware.
- Because data is on the cloud, there can be no single point of failure.
- Enables the virtualization of administrative tasks, freeing up time for other work.

- **Example:** Amazon Web Services (AWS), Microsoft Azure, Google Compute Engine (GCE), Rackspace, Cisco Metacloud.

PLATFORM AS A SERVICE

PaaS cloud computing platform is created for the programmer to develop, test, run, and manage the applications.

- **Key features:**
- PaaS provides a platform with tools to test, develop, and host applications in the same environment.
- Enables organizations to focus on development without having to worry about underlying infrastructure.
- Providers manage security, operating systems, server software and backups.
- Facilitates collaborative work even if teams work remotely.

- **Example:** Windows Azure, Google App Engine, Apache Stratos.

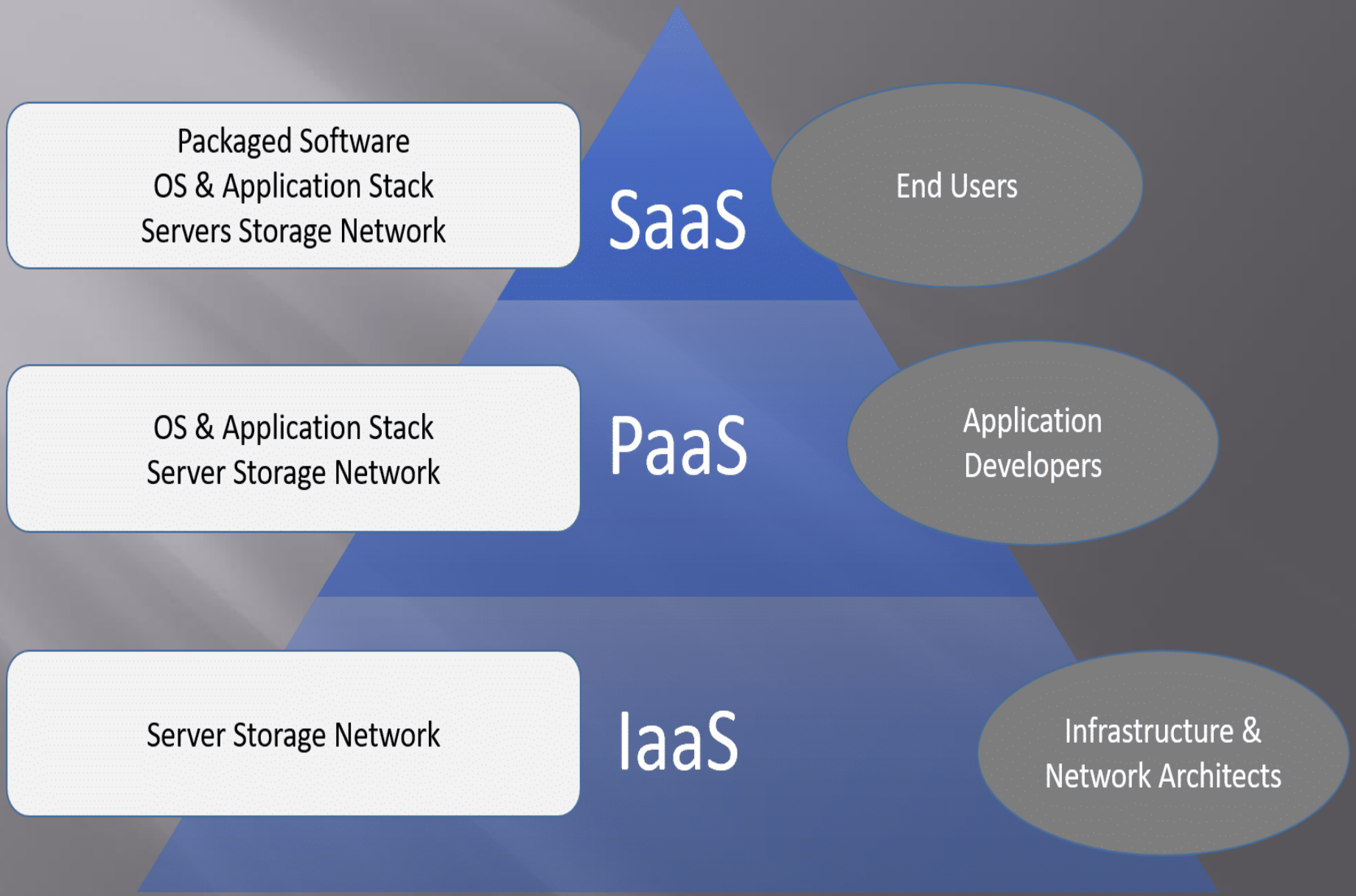
SOFTWARE AS A SERVICE

SaaS is also known as "**on-demand software**". It is a software in which the applications are hosted by a cloud service provider. Users can access these applications with the help of internet connection and web browser.

- **Key features:**
- SaaS vendors provide users with software and applications via a subscription model.
- Users do not have to manage, install or upgrade software; SaaS providers manage this.
- Data is secure in the cloud; equipment failure does not result in loss of data.
- Use of resources can be scaled depending on service needs.
- Applications are accessible from almost any internet-connected device, from virtually anywhere in the world.

- **Example:** BGoogle Apps, Salesforce, Dropbox, ZenDesk, Cisco WebEx.

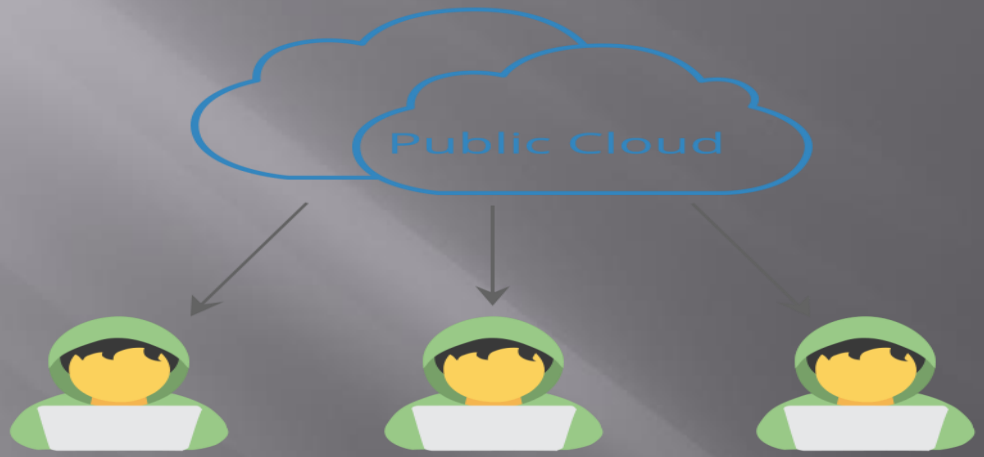
Cloud Service Models



PUBLIC CLOUD DEPLOYMENT

- While the name speaks for itself, a public cloud is a cloud where services are rendered by third-party providers over an open network for public use.
- Yes. It's available to the general public and data is created & stored on third-party servers.
- The public cloud deployment model is one of the most commonly used cloud services. It's a popular option for apps, file sharing, and insensitive data storage.
- Under this model, the service provider owns and operates all the hardware that is needed to run a public cloud.

PUBLIC CLOUD



They keep devices in massive data centers. This model also plays a vital role in development and testing. That's because it's cheap, easily configurable, can be deployed quickly, and supports fluctuating demands

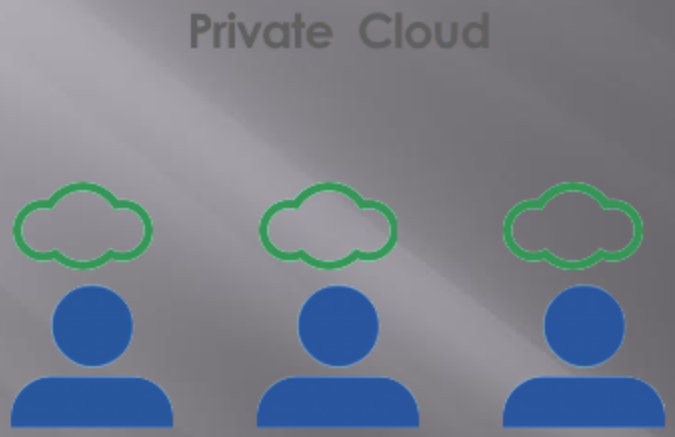
Advantages

- **Low Cost** – The service providers are the ones who fund the entire infrastructure (no need for you to invest in hardware or software). And aside from the initial fee, you're good to go on the 'pay per use' policy under this model, which means, no overhead costs.
- **Convenient Infrastructure Management** – You don't need to develop or maintain your software because the service provider does that for you. The setup and use are convenient.
- **Time** – It reduces time in developing, testing, and launching new products. Aside from that, the extensive network of your provider's servers ensures that your infrastructure is available for use 24X7.

Disadvantages

- **Data and Security Risk** – As the name suggests, it's a public platform for insensitive data storage. As a result of shared resources, security risks are high due to their vulnerability. A user is deprived of knowing where their information is being stored and who all have the access to it.
- **Network Performance** – Due to an increase in the number of users, the network performance suffers from instability.
- **Lack of Customization** – Since it's a public cloud, personalization will always be limited.

Private Cloud Deployment



A private cloud deployment model, also known as an internal or corporate model, is one that belongs to specific organizations. Under this model, the organization is the one that controls the systems and manages them. And while a third party (a service provider) can host a private cloud server, most companies choose to keep their hardware in their on-premises data center from where an in-house team oversees and manages everything. The private cloud deployment model is the second most used model and has it's own pros and cons which we've stated below:

PRIVATE CLOUD ADVANTAGES

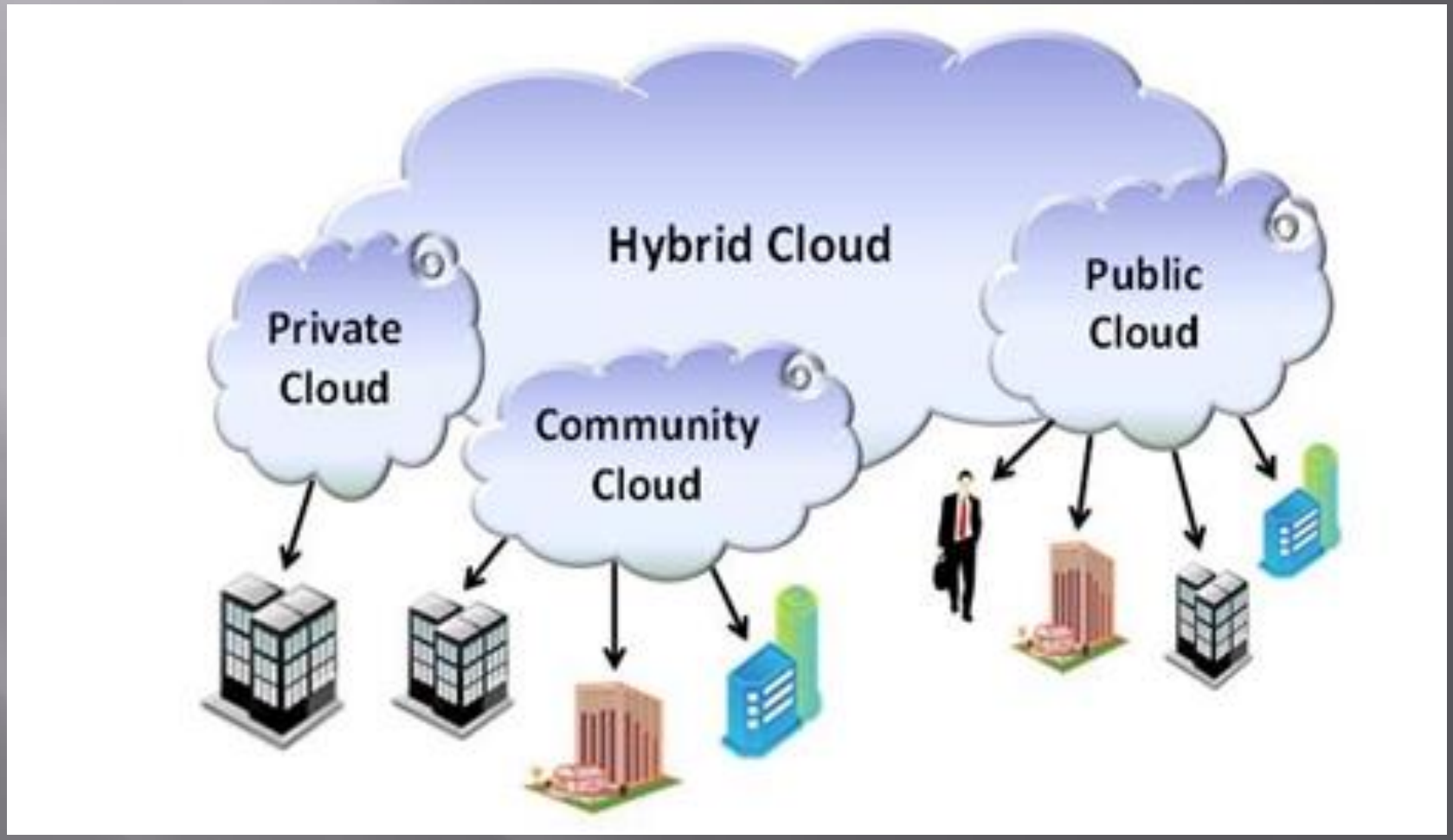
Advantages

- **Customization** – Unlike the public cloud deployment model, a company using a private cloud can customize its solution as per its own requirements.
- **Data and Security** – The security of data is high, as in, only authorized internal personnel is allowed to access data. This makes it ideal for storing corporate data. A company can also separate the sets of resources on the same infrastructure which makes it even more secure.
- **Control** – Your organization is the exclusive owner. You get absolute control over service integrations, IT operations, rules, and user practices.
- **Legacy Systems** – A special feature about private clouds is that it supports legacy applications that are not functional on public clouds.

Disadvantages

- **High Cost** – Yes! You'll need to invest in hardware and software. Aside from that, there are also expenses for in-house staff and training. And not to forget maintenance, which is quite high.
- **Limited Scalability** – Private clouds are scaled within internal limited hosted resources. Scalability also depends on the choice of your underlying hardware.

HYBRID CLOUD



SOFTWARE AS A SERVICE

A hybrid cloud deployment model is a combination of two or more cloud servers as one architecture. With a mix of two or more cloud deployment models, organizations are capable of moving data and applications between different clouds, depending on their purpose.

HYBRID CLOUD ADVANTAGES

- **Benefits of Hybrid Cloud**
- Cost-Effectiveness - The overall cost of a hybrid solution decreases since it majorly uses the public cloud to store data.
- Security - Since data is properly segmented, the chances of data theft from attackers are significantly reduced.
- Flexibility - With higher levels of flexibility, businesses can create custom solutions that fit their exact requirements
- **Limitations of Hybrid Cloud**
- Complexity - It is complex setting up a hybrid cloud since it needs to integrate two or more cloud architectures
- Specific Use Case - This model makes more sense for organizations that have multiple use cases or need to separate critical and sensitive data
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COMMUNITY CLOUD



A community cloud deployment model resembles a private cloud, the only difference being the set of users. While a private cloud implies that one company owns the server, in the case of a community cloud, it is several organizations with similar backgrounds. Companies share the infrastructure and related resources on a community cloud. This model supports joint business organizations, ventures, research organizations, and tenders, etc. Community Cloud is a way to preserve the benefits of economy of scales with the private cloud.

COMMUNITY CLOUD ADVANTAGES

- **Advantages**
- • **Cost –** A community cloud is a cheaper way to avail the benefits of a private cloud. That's because multiple companies can share the bill together.
  • **Data Sharing and Collaboration –** With a community cloud, sharing data becomes easy and its collaborative space allows clients to enhance their efficiency. Aside from that, configuration and protocols within a community system meet the needs of the specific industry.
- **Disadvantages**
- • **Limited Storage and Bandwidth–** Within community systems, limited storage and bandwidth capacity become a common problem.

THANK YOU